

Vishal Indivar Kandala

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RESEARCH PROFILE

Ph.D. Candidate in Mechanical Engineering with a focus on computational science, viewing computation as a third pillar of scientific inquiry alongside theory and experiment. My research centers on developing bespoke, high-fidelity software tools from first principles to investigate complex physical phenomena. My current Ph.D. work involves building **PICurv**, a parallel Eulerian-Lagrangian solver for data-driven turbulence modeling. Previous projects have applied these computational methods to diverse applications, including biomedical fluid-structure interaction in LVADs and physics-informed modeling of cardiac dynamics (EDPVR). This work is driven by a deep appreciation for the predictive power of computational physics, which began with hands-on experience in undergraduate engineering competitions and has grown through tackling progressively complex research challenges.

EDUCATION

2021 - present	Ph.D., Mechanical Engineering at Texas A&M University Advisor: Dr. Iman Borazjani Dissertation: Data-Driven Turbulence Closure Models using Filtered Density Function Methods	College Station, TX
2019 - 2021	M.Eng., Aerospace Engineering at Texas A&M University College Station, TX	(GPA: 3.8/4.0)
2014 - 2018	B.Tech., Mechanical Engineering at VIT University Chennai, India	(GPA: 8.0/10.0)

RESEARCH INTERESTS

Primary Focus	Turbulence modeling, Data-driven closure development, Filtered Density Function (FDF) methods, Large Eddy Simulation (LES)
Computational Methods	High-performance computing, Eulerian-Lagrangian methods, Immersed boundary methods, Parallel computing, Numerical methods for PDEs
Application Areas	Turbulent scalar transport, Combustion modeling, Fluid-structure interaction, Biomedical flows, Particle-laden flows
Emerging Interests	Physics-informed machine learning, Scientific machine learning for turbulence, Hybrid numerical-ML approaches

RESEARCH EXPERIENCE

Graduate Research Assistant — Texas A&M University Aug 2021 - Present
Scientific Computing and Biofluids Lab, Advisor: Dr. Iman Borazjani

- PICurv: Eulerian-Lagrangian Solver for Turbulent Scalar Transport** Jan 2024 - Present
- Developing a high-fidelity solver for scalar **Filtered Density Function (FDF)** transport within the **CURVIB** immersed boundary framework.
 - Implementing Eulerian-Lagrangian coupling and solving stochastic particle methods (Langevin equation) to model sub-grid scalar fluctuations in turbulent flows.
 - Building a modular, **well-documented**, **open-source tool** with a focus on robust software engineering for the scientific community.

- Optimizing parallel scalability to handle millions of particles across 1,000+ cores using C/C++, MPI, and PETSc.
- This work establishes the foundation for a future [velocity-scalar FDF \(VFDF\)](#) framework, enabling a new class of data-driven turbulence closure models.

Biomedical Computational Mechanics Projects

May 2021 - Dec 2023

Left Ventricular Assist Device (LVAD) Flow Analysis:

- Conducted high-resolution, unsteady fluid-structure interaction simulations of pulsatile blood flow within patient-specific LVAD outflow graft geometries.
- Analyzed key hemodynamic metrics including **wall shear stress**, vorticity, and regions of flow stasis to quantify the risk of thrombosis (blood clotting) and hemolysis (blood cell damage).
- Utilized the Curvilinear Immersed Boundary (CURVIB) method to accurately capture the interaction between blood flow and the flexible arterial walls.
- Presented a detailed analysis of the outflow graft hemodynamics at the 76th Annual APS Division of Fluid Dynamics meeting.

Cardiac Pressure-Volume Modeling (EDPVR):

- In a cross-disciplinary collaboration with the [Cardiovascular and Computational Biomechanics Lab](#), developed a novel framework for modeling right ventricular pressure-volume dynamics.
- Designed and implemented a **physics-informed regression model** that extends the established Klotz single-beat method to the anatomically complex right ventricle.
- Generated training data by performing finite element simulations (Abaqus) on a large cohort of biventricular geometries reconstructed from high-resolution MRI scans.
- Demonstrated a **40-60% reduction in prediction error** (nRMSE reduced from 22.7% to 12.3%) by introducing a geometric correction factor, enabling more accurate, non-invasive cardiac assessment. Manuscript in preparation for *Nature Scientific Reports*.

Undergraduate Research Assistant — VIT University

Jun 2017 - May 2018

Advisor: Dr. [Harish Rajan](#)

Hypersonic Reentry Vehicle Design Optimization

- Conducted a design optimization study for a modified double-spiked reentry capsule using ANSYS Fluent to mitigate aerothermal loads.
- Performed parameter sweeps varying aerospike frustum geometry, achieving a 25% drag reduction and 16% surface temperature reduction for the optimized configuration.
- Authored a technical report on the findings, demonstrating a systematic approach to design space exploration through computational simulation.

PUBLICATIONS & PRESENTATIONS

Conference Presentations

Kandala, V.I. and Borazjani, I.

2025

"A Parallel Particle-In-Cell (PIC) Solver on Curvilinear Grids for Turbulent Flow Simulation," *78th Annual APS Division of Fluid Dynamics*, Houston, TX. (Accepted) [[Schedule](#)]

Kandala, V.I. and Borazjani, I.

2023

"High Resolution Numerical Simulations of LVAD Outflow Graft Haemodynamics," *76th Annual APS Division of Fluid Dynamics*, Washington, DC. [[Abstract](#)] [[Presentation](#)]

Posters

Naeini, V., Kandala, V., et al.

2023

"On the non-invasive estimation of end diastolic pressure volume relationship in the right ventricle," *BMES Annual Meeting*, Seattle, WA. [[Abstract](#)]

Kandala, V.I. and Borazjani, I.

2022

"High Performance Computing for Biofluids," *High Performance Research Computing Symposium*, Texas A&M University. [[Poster](#)] [[Event](#)]

Manuscripts in Preparation

Naeini, V., **Kandala, V.**, Mukherjee, T., Mehdi, R.R., Borazjani, I., Choudhary, G., and Avazmohammadi,

"In-silico-generated universal end-diastolic pressure-volume relationship for the right ventricle," *in preparation for Nature Scientific Reports*.

Undergraduate Publications

Kandala, V.I. and Kumar, S.

2018

"Numerical Study of a Modified Double-Spiked Re-entry Capsule," VIT University. [[PDF](#)]

TEACHING EXPERIENCE

Teaching Assistant — Texas A&M University

Aug 2019 - May 2024

STAT 201: Statistical Methods

Fall 2019, Spring 2020, Fall 2020

- Instructed 200+ students from diverse academic backgrounds in statistical inference
- Developed materials emphasizing practical applications and conceptual understanding

MEEN 461: Heat Transfer

Spring 2021, Fall 2021, Spring 2022

- Taught graduate students and senior undergraduates from ME, AE, and ChemE programs
- Created assignments, projects, and exams; delivered lectures; conducted tutorials using MATLAB and Python

MEEN 464: Heat Transfer Laboratory

Fall 2022, Spring 2023, Fall 2023, Spring 2024

- Supervised small teams (4-5 students) conducting heat transfer experiments
- Developed lab manuals and experimental protocols; managed laboratory equipment

PROFESSIONAL EXPERIENCE

Product Development Intern — EPC Electrical Pvt. Ltd.

Jan 2018 - Aug 2018

- Co-led the end-to-end design, fabrication, and deployment of a novel, ultra-low-cost CNC machine, achieving a time-to-market of under 90 days.
- Reduced customer operating costs by 60% by implementing the GRBL open-source firmware and control platform.

R&D Engineering Intern — BHEL

May 2017 - Aug 2017

- Selected as one of 20 interns from a pool of 300+ applicants to contribute to R&D at India's largest energy equipment producer.
- Developed a MATLAB application to compute heat transfer coefficients in industrial condensers, eliminating commercial software licenses and saving an estimated \$2,500 annually.

INDEPENDENT PROJECTS

Cantera-Radcal: Chemical Kinetics with Radiation

Mar 2020 - Jan 2021

- Extended the open-source [Cantera package](#) to include radiation modeling and higher-order reverse diffusion as part of my master's thesis project.
- Integrated the modified package with a compressible OpenFOAM solver to simulate non-premixed flames, a critical step for high-speed combustion applications.

Foundational CFD Solver Implementations

Coursework

Self-Study

- Developed a portfolio of solvers from first principles to build a fundamental understanding of numerical methods and solver architecture.
- **Parallel Diffusion Solver:** A 2D Finite Difference solver in C using the PETSc library for parallel execution and scalability analysis. [[GitHub](#)]
- **Artificial Compressibility Solver:** A 2D Navier-Stokes solver for backward-facing step flow using an explicit Runge-Kutta time-marching scheme on a stretched grid. [[GitHub](#)]
- **SIMPLE Algorithm Solver:** A 2D Finite Volume solver in Python implementing the SIMPLE algorithm on a staggered grid for channel and lid-driven cavity flows. [[GitHub](#)]

OPEN SOURCE SOFTWARE

PICurv

[GitHub](#) — [Documentation](#)

High-fidelity Eulerian-Lagrangian solver for particle-laden turbulent flows on curvilinear grids. Features modular design, comprehensive Doxygen documentation, automated test cases, and parallel scalability.

Cantera-Radcal

[GitHub](#)

Extension of Cantera chemical kinetics package with radiation modeling (RADCAL) and higher-order diffusion. Docker implementation for reproducibility.

Parallel Diffusion Equation Solver

[GitHub](#)

2D diffusion equation solver using finite difference method with PETSc library's C interface. Features Krylov subspace methods for efficient parallel solution.

Py-SIMPLE

[GitHub](#)

2D finite volume method solver for incompressible, steady-state flows written in Python with NumPy. Implements SIMPLE algorithm for pressure-velocity coupling.

Parallel Artificial Incompressibility Solver

[GitHub](#)

High-performance CFD solver for 2D steady-state incompressible Navier-Stokes equations on generalized curvilinear grids using artificial incompressibility method.

1D Composite Diffusion Solver

[GitHub](#)

Simulates 1D heat conduction in spherical bodies with multiple layers of different material compositions and varying thicknesses in the radial direction.

TECHNICAL SKILLS

Programming & HPC	C/C++, Python, MATLAB, MPI, OpenMP, PETSc, CUDA (exposure)
CFD & Simulation	ANSYS Fluent, OpenFOAM, STAR-CCM+ (exposure), Converge (exposure), Cantera
Numerical Methods	Finite Volume Method, Finite Element Method, Immersed Boundary Methods, Spectral Methods, Time Integration (RK, fractional-step), Iterative Solvers (GMRES, Krylov)
Meshing & Pre-processing	ICEM CFD, Pointwise, Gmsh, ANSYS Meshing
Post-processing	Tecplot, ParaView, EnSight, Matplotlib
CAD	SolidWorks, Creo, SpaceClaim
HPC Systems	National supercomputing resources, cluster management, job scheduling
Version Control	Git, GitHub

AWARDS & HONORS

2022	Chad Patel Scholarship , American Society of Indian Engineers
2019, 2020	Aerospace Academic Excellence Fellowship , Texas A&M University

PROFESSIONAL SERVICE & LEADERSHIP

Judge — Texas Science Fair

2023

Evaluated student research projects and provided feedback to young researchers

Department Representative — TAMU Graduate and Professional Student Government

Represented mechanical engineering graduate students in university governance

Contributing Writer — Ingenium Blog, Texas A&M Engineering

2020 - 2022

Wrote articles about graduate student experiences and research for university engineering blog

Co-Founder & Head of Aerodynamics — Team Aviators International

2016 - 2017

– Founded first SAE Aero Design team at VIT University; led 5-member aerodynamics sub-team

– Secured \$5,000+ in funding through workshops and sponsorships

Lead Braking Engineer — Team Saksham (Baja SAE India)

2015 - 2016

– Designed and fabricated braking system; achieved 5th place nationally (100+ teams)

– Contributed to fundraising efforts securing \$3,000+ for components and travel

RELEVANT COURSEWORK

CFD & Turbulence Modeling

Turbulence Modeling (AERO 670), Turbulence Processes (OCEN 640), Computational Fluid Dynamics (MEEN 689)

Propulsion & High-Speed Flows

High-Speed Combustion & Propulsion (AERO 641), Aerothermochemistry (AERO 676), Advanced Aerodynamics (AERO 601)

Scientific & Parallel Computing

Scientific Machine Learning (ECEN 689), Parallel Computing (CSCE 689), Advanced Computer-Aided Engineering (MEEN 632)

Numerical Methods & Analysis

Numerical Heat Transfer & Fluid Flow (MEEN 644), Numerical Methods in Differential Equations (MATH 610), Numerical Analysis (AERO 689)

Fundamental Mechanics

Theory of Fluid Mechanics (AERO 602), Fluid Mechanics (MEEN 621), Introduction to Random Dynamical Systems (AERO 630)

PROFESSIONAL MEMBERSHIPS

American Physical Society (APS) – Division of Fluid Dynamics

REFERENCES

Dr. Iman Borazjani

Ph.D. Advisor

Professor, J. Mike Walker '66 Department of Mechanical Engineering

Texas A&M University — Email: iman@tamu.edu

Dr. Reza Avazmohammadi

Research Collaborator

Associate Professor, Department of Biomedical Engineering

Texas A&M University — Email: rezaavaz@tamu.edu

Dr. Harish Rajan

Undergraduate Research Advisor

Associate Professor, School of Mechanical Engineering

Vellore Institute of Technology (VIT), Chennai — Email: harish.r@vit.ac.in